

# *E9 205 Machine Learning for Signal Processing*

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## Dimensionality Reduction - II

30-08-2017

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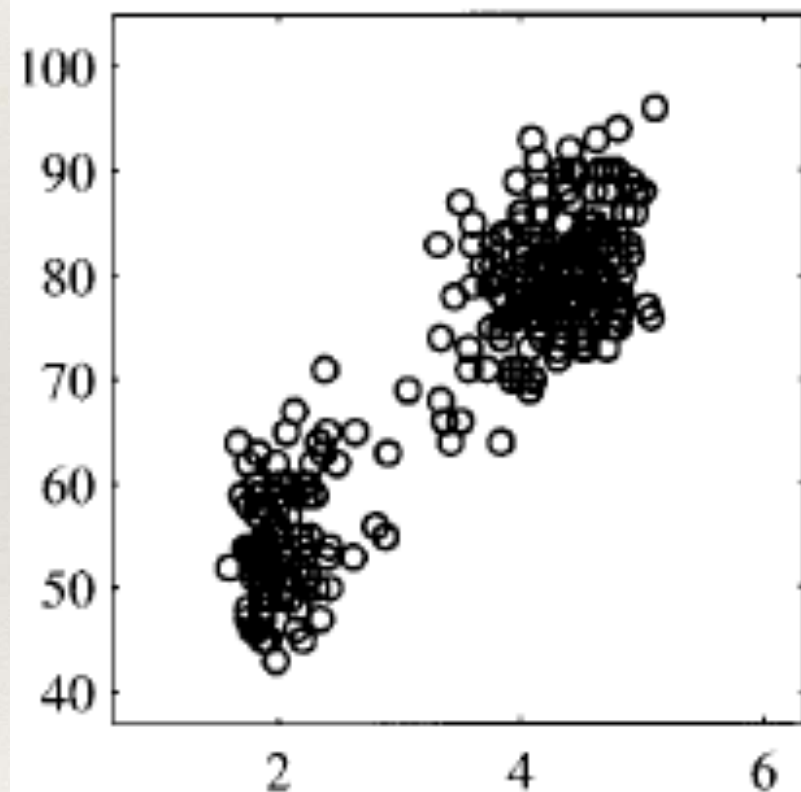
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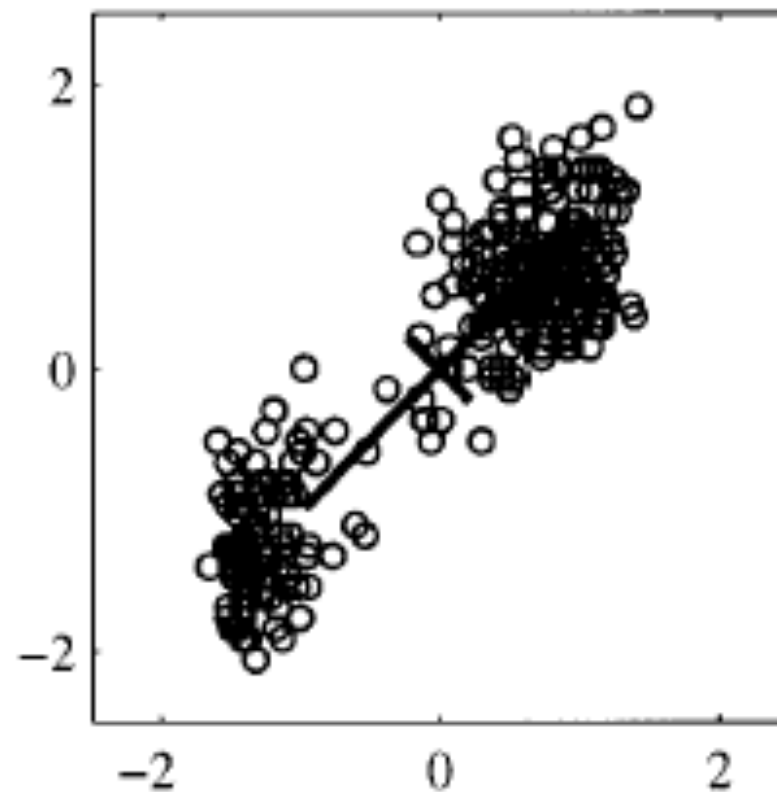


# Whitening the Data

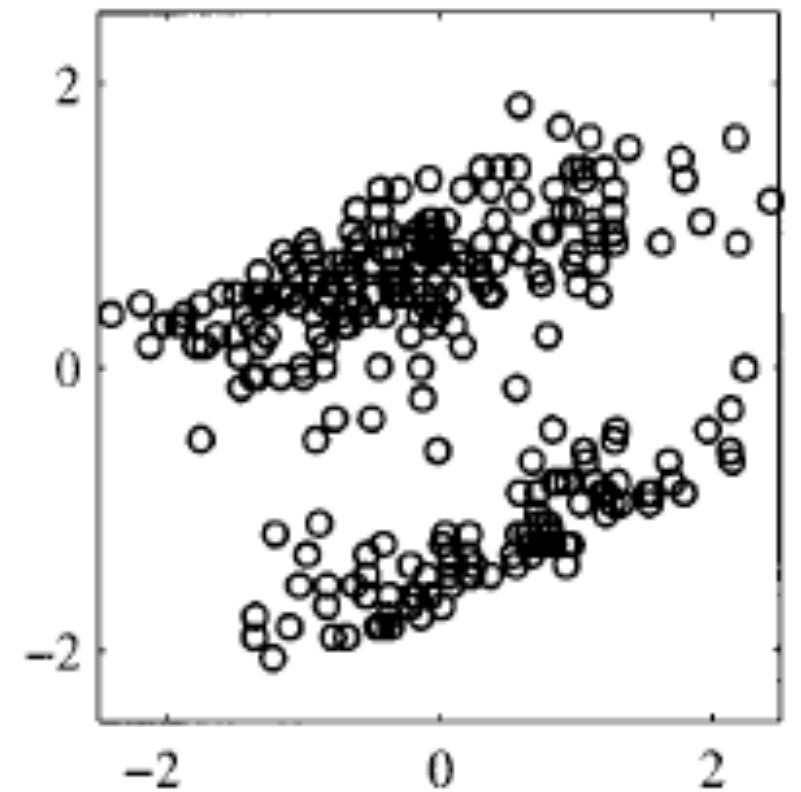
Original Data



Whitened data

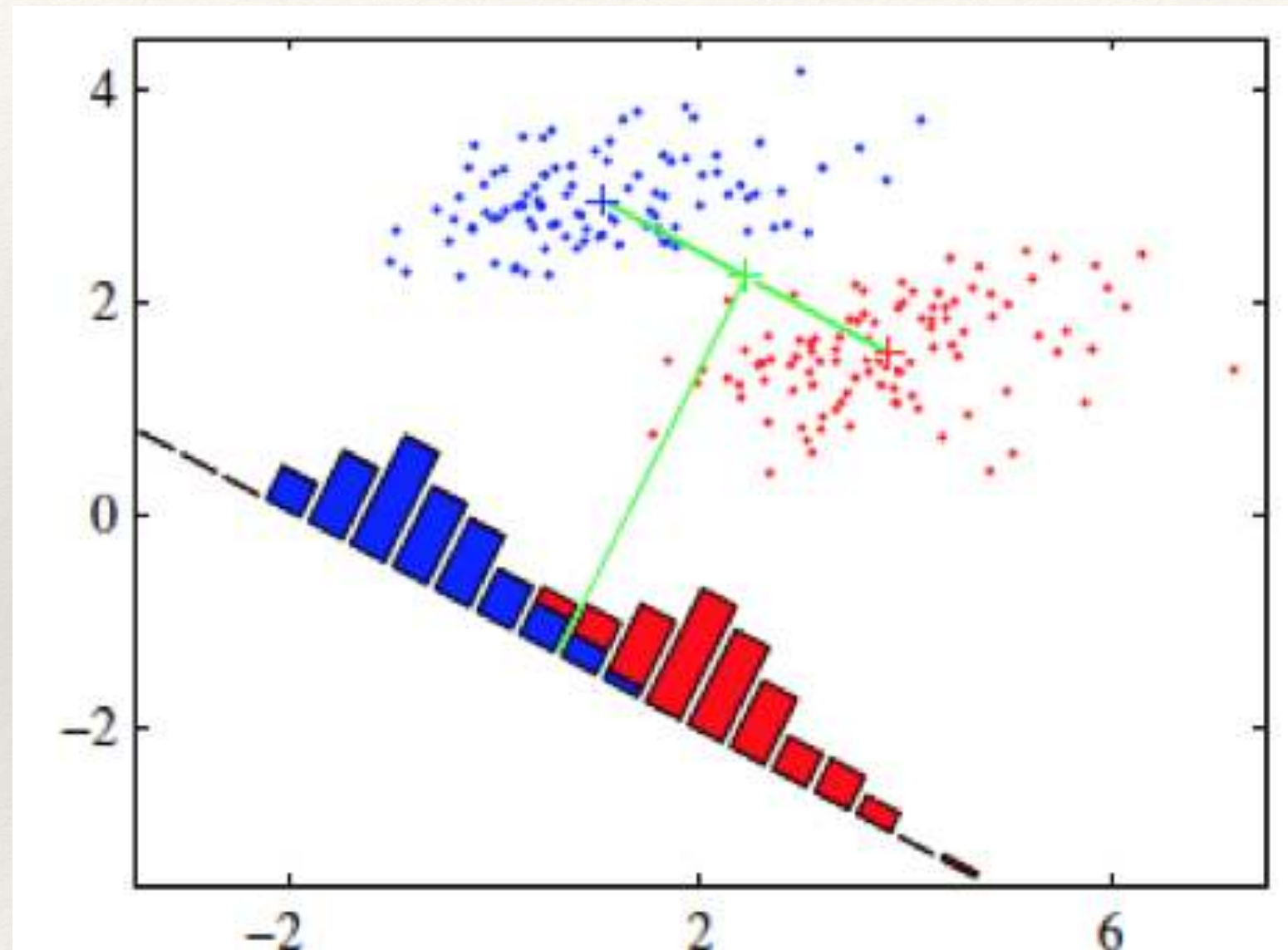


Whitened data





# With Only the Within Class Factor



# Linear Discriminant Analysis

Find a linear transform  $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$  with a criterion which maximizes the class separation

- Maximize the between class distance in the projected space while minimizing the within class covariance

$$J = \frac{\mathbf{w}^T \mathbf{S}_b \mathbf{w}}{\mathbf{w}^T \mathbf{S}_w \mathbf{w}}$$

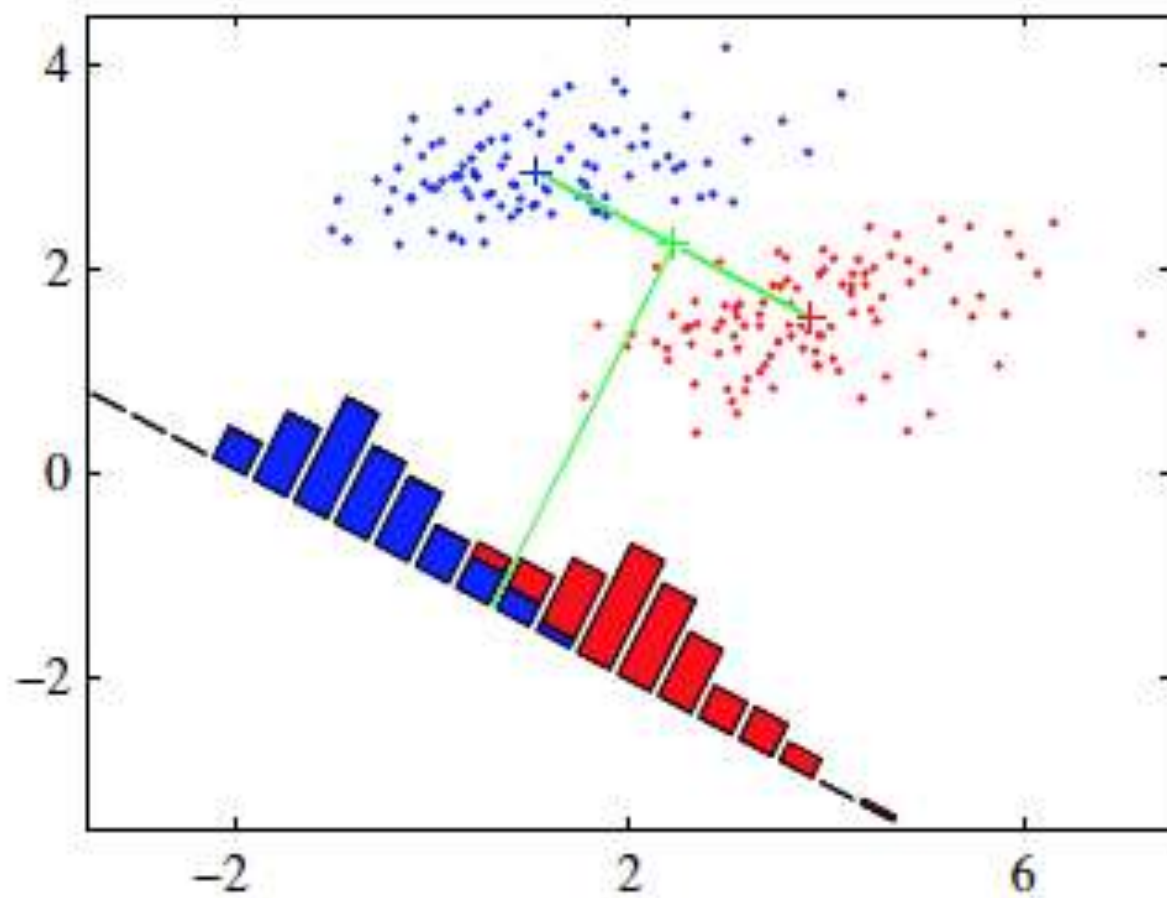
$$\mathbf{S}_b = \sum_{k=1}^K N_k (\mathbf{m}_k - \mathbf{m})(\mathbf{m}_k - \mathbf{m})^T \quad \mathbf{S}_w = \sum_{k=1}^K \sum_{n \in C_k} (\mathbf{x}_n - \mathbf{m}_k)(\mathbf{x}_n - \mathbf{m}_k)^T$$

- ❖ Generalized Eigenvalue problem
- ❖ Eigenvectors of  $\mathbf{S}_w^{-1} \mathbf{S}_b$

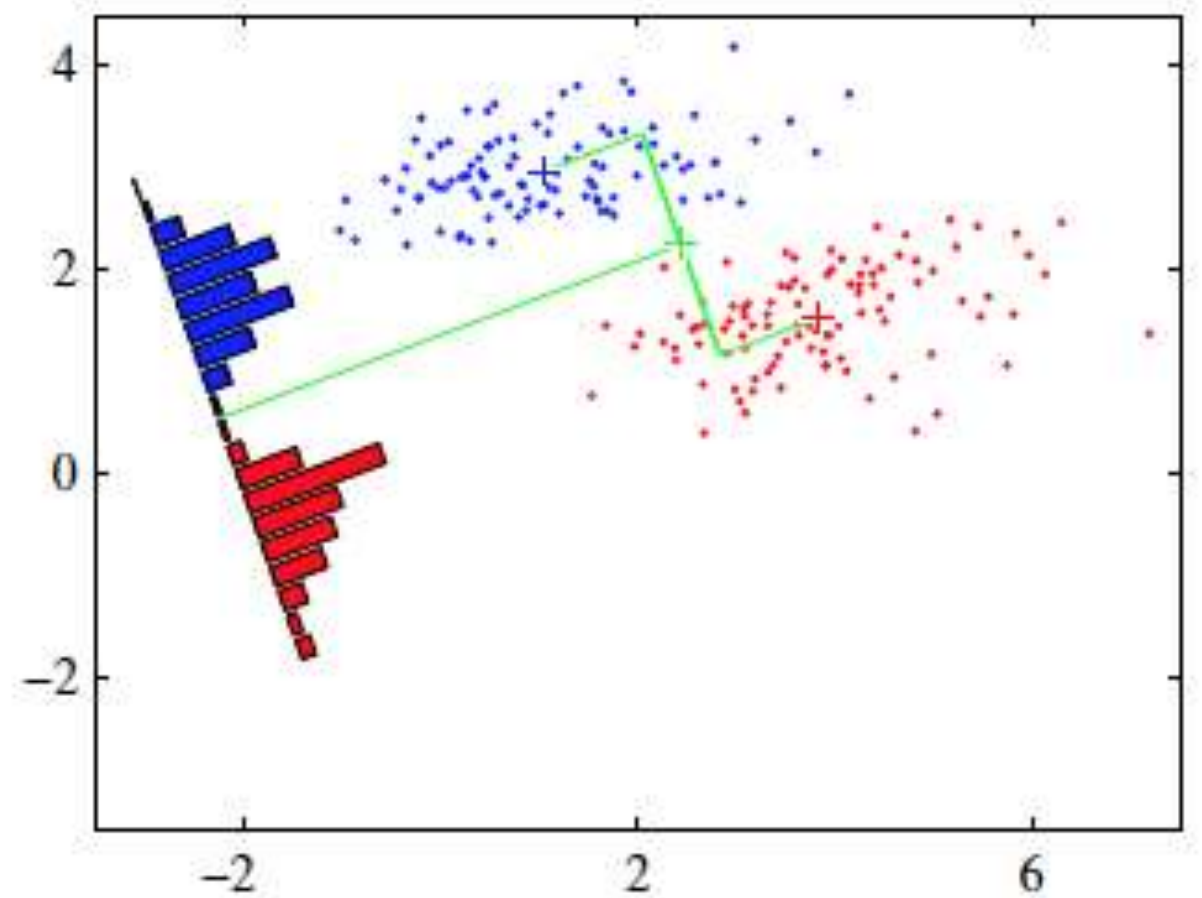


# Linear Discriminant Analysis

Projecting on line joining means



Fisher Discriminant



# PCA versus LDA

